

Data Flow Diagram

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
External Entities	Farmer / Production Planner	Sends raw soil telemetry metrics (N, P,K, soil temperature, humidity, pH, rainfall) via the front-end interface and receives crop recommendations.
Processes	Input Validation AI Analytics & Predictive Modeling	Receives raw form input from the user and ensures all 7 numerical agricultural metrics fall within realistic physiological bounds. Handled by Jaheer Vali Sheik and syamala perapu as important in team.
Data Stores	soil_telemetry_metrics	Holds historical data entries, metric logs,calculated crop predictions, and confidence levels generated by the core analytics engine.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

